

on chips (SoCs), field programmable gate arrays (FPGAs), and application specific integrated circuits (ASICs) mostly come in ball grid array (BGA) packages, some ICs are available in multiple packages like SSOP, TSSOP, SO, etc. The key considerations to select a package are cost, temperature range, heat dissipation, real estate on the PCB, and lead time.

After the parts are finalised, symbols and footprints corresponding to those parts need to be created to proceed with the schematic. It is important to review the design carefully to check the desired functionality and any mistakes like wrong connection, unconnected nets, incorrect package selection, etc.

Component footprints also need to be checked carefully for their mechanical dimensions, pin sequences, etc. Only after that the final netlist and the bill of material (BoM) should be generated.

PCB stack-up and board mechanical. The PCB stack-up decides which layer would be used for signal routing and which ones for power and ground planes. The number of layers required for a particular design may be decided based on requirements, recommendations of the device datasheets, and the designer's experience.

The board mechanical, or PCB template, is an auto-CAD drawing that specifies the shape and dimensions of the PCB, coordinates of the components sitting at the edge of the PCB (mostly connectors, LEDs, switches, jacks, etc), mounting holes, and other constraints that might be specific to the design.

The template may require minor modifications after the thermal simulation is done, as additional mounting holes may be required to fix heat sinks, if required.

Thermal simulation. During thermal simulation of the PCB, rough placement of the major heat-

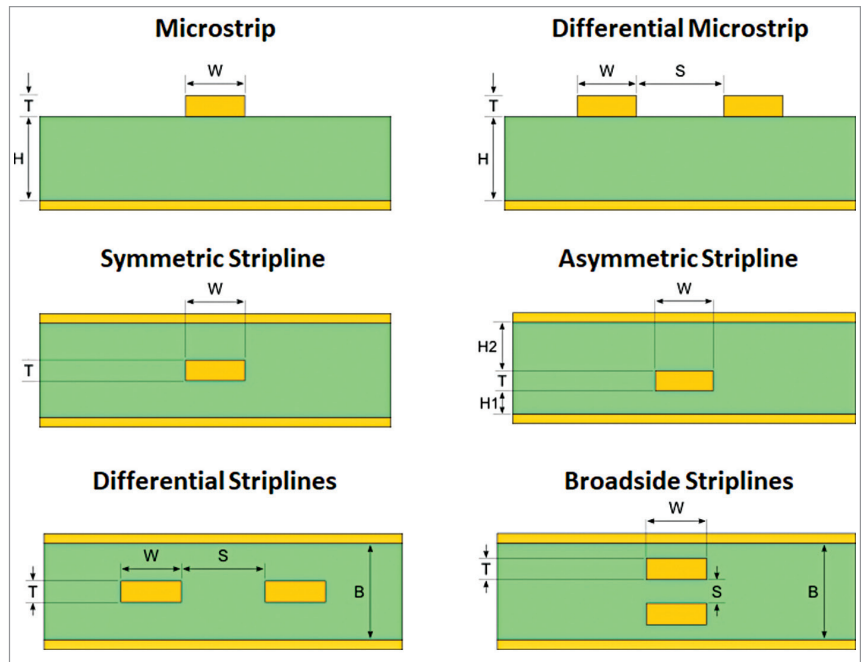


Fig. 2: Types of microstrip and stripline (Credit: <https://resources.altium.com>)

generating ICs, power supplies, components with special features (for example, a tall capacitor that might be blocking airflow), and ambient operating temperature need to be specified. Thermal inputs for an IC include total heat dissipation, maximum junction temperature, and thermal resistance parameters.

For pluggable modules in cages, the thermal model is required for accurate simulation. Thermal simulation may recommend modifying the placement, if required. Also, the simulation decides the specification of fans, if required for cooling.

Placement and routing. One of the most critical and time-consuming steps in high-speed PCB design is component placement and routing. PCB layout is nothing but a multivariable optimisation problem, the solution to which just gets better with experience. PCB layout includes defining constraints like keep-outs (where routing and/or planes are not allowed), placement of the components, routing

of signals, creation of power and ground planes, optimisation of routes, removal of unwanted vias, and cleaning of designs from extra traces or stubs.

In complex high-speed PCBs, signal integrity (SI) analysis (explained in next step) is a crucial step. It is generally done in two phases—Pre SI (before routing) and Post-SI (after routing). Also, we need to perform Design For Manufacturing (DFM) or Design For Assembly (DFA) checks to avoid any last-minute issues.

After the routing, its review, and SI analysis are done, the next step is back annotation where the components on the PCB are renumbered in sequential order based on their location on the PCB. This process helps in finding any component that may have been missed during debugging and board-level hardware testing.

SI analysis. SI (signal integrity) is a key challenge for the circuit to function properly. Many PCB design efforts fail just because signal integrity is not maintained.